

Lecture 2

Array Basics



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Objective



Learn basic operation of Array

Understand How it works

The system of doing operation

Learn through Movie ← ←

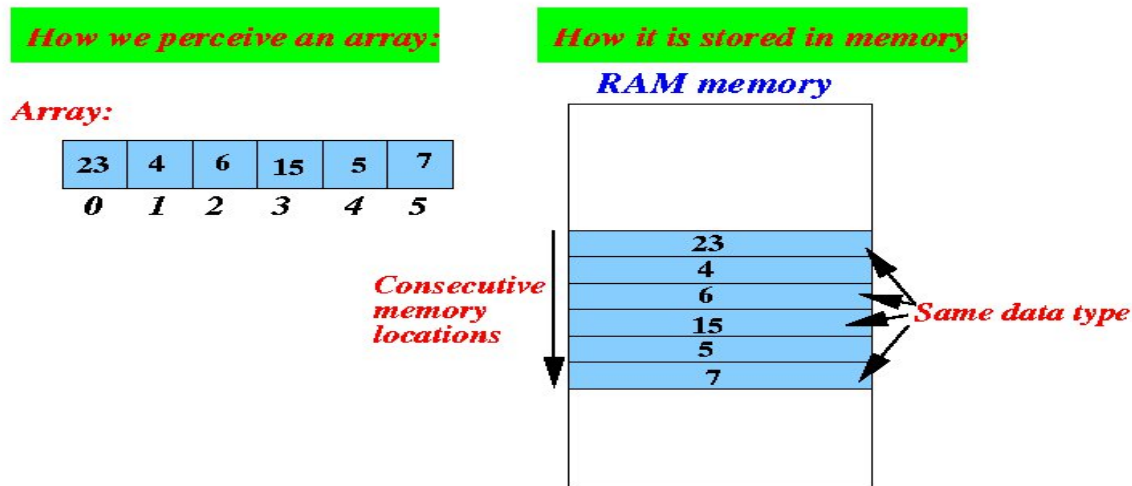
Array

Linear array (One dimensional array) : A list of finite number n of similar data elements, usually $1, 2, 3, \dots, n$.

Let, Array name is A , has 6 elements. Then the **values** of the elements of A is : $23, 4, 6, 15, 5, 7$ i.e. $A[6]=[23, 4, 6, 15, 5, 7]$

Index, k : $0, 1, 2, 3, 4, 5$.

The number k in $A[k]$ is called a subscript and $A[k]$ is called a subscripted variable. So, $A[0]=23, A[1]=4 \dots$ etc.



Array (con...)



Two Dimensional Array : Two dimensional array is a collection of similar data elements where each element is referenced by two subscripts.

Such arrays are called matrices in mathematics and tables in business applications.

Example : Array A.

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

	Column 0	Column 1	Column 2	Column 3
Row 0	a[0][0]	a[0][1]	a[0][2]	a[0][3]
Row 1	a[1][0]	a[1][1]	a[1][2]	a[1][3]
Row 2	a[2][0]	a[2][1]	a[2][2]	a[2][3]

Here, $A[2,3]=12$

Array Data Structure



- ❑ It can hold multiple values of a single type.
- ❑ Elements are referenced by the array name and an index.
- ❑ Each element is a *value*
- ❑ Indexing begins at zero.

Array Operations



Traversing

Visiting every element once of an array and do some work such as printing every element, calculate total element etc.

Say, $A[3] = \{10, 15, 20\}$

For printing this array we need to visit this array element once.

For Adding all element or counting total element of this array we also need to visit this array element once.

Array Operations



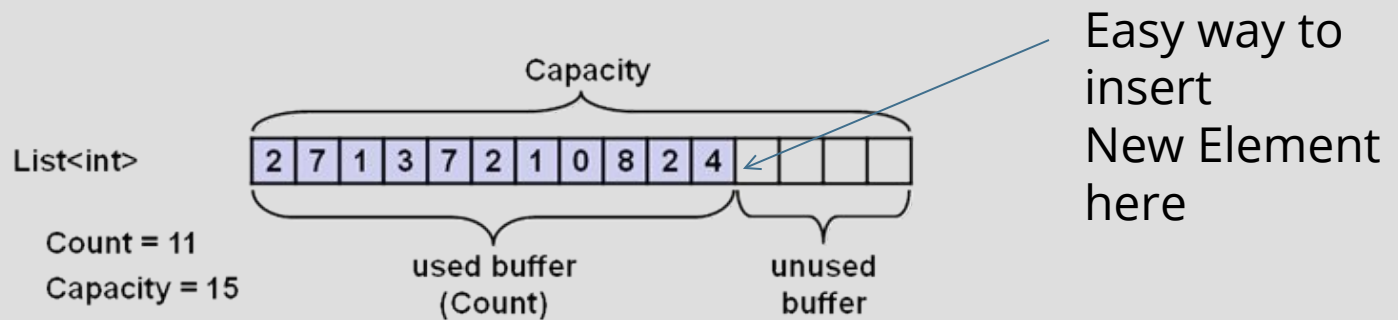
Inserting

- In inserting operation , new data items are added or inserted into any location of an empty or filled array.
- In case of empty array, new values are stored in empty locations.
- In case of filled array , old values are either replaced by new values or pushed backward to make room for new values.
- It is very easy to insert data at the end of array(Why?).
- Without disturbing the data of other elements of the array.

Array Operations



Suppose an array $a[15]=\{2,7,1,3,7,2,1,0,8,2,4\}$



Now, If we need to insert data in 2nd position then?

Need to insert new value in proper position and then array will be

$A[15]=\{2, \text{New Value}, 7, 1, 3, 7, 2, 1, 0, 8, 2, 4\}$

Array Operations



Deletion

- Deleting item from a specified Location
- When the data item is removed from a specified location within an array, the items from that location up to the end of array are moved one location towards the beginning of the array.

Let array $A = \{1, 2, 3, 4, 5, 6\}$

Want to delete 3
OR
value of third position



Output will be:



Array Operations



Searching : Linear search

DATA is a linear array with n elements. The most intuitive way to search for a given ITEM in DATA is to compare ITEM with each element of DATA one by one. That is first we test whether $\text{DATA}[1] = \text{ITEM}$, and then we test whether $\text{DATA}[2] = \text{ITEM}$, and so on. This method, which traverses DATA sequentially to locate ITEM, is called **linear search** or **sequential search**.

We want to find where 3 is situated, Or whether 3 is present or not? Then, Check Data[0] with 3. If not match then check with DATA[1] and so on.

Key	List
3	[6] 4 1 9 7 3 2 8
3	[6] [4] 1 9 7 3 2 8
3	[6] 4 [1] 9 7 3 2 8
3	[6] 4 1 [9] 7 3 2 8
3	[6] 4 1 9 [7] 3 2 8
3	[6] 4 1 9 7 [3] 2 8
3	[6] 4 1 9 7 3 [2] 8
3	[6] 4 1 9 7 3 2 [8]

Array Operations



Searching: Binary Search

If we place our items in an array and sort them in either ascending or descending order on the key first, then we can obtain much better performance with an algorithm called **binary search**. In binary search, we first compare the key with the item in the middle position of the array. If there's a match, we can return immediately. If the key is less than the middle key, then the item sought must lie in the lower half of the array; if it's greater then the item sought must lie in the upper half of the array. So we repeat the procedure on the lower (or upper) half of the array.

Array Operations



Say array $a[10]=\{1,2,3,9,11,13,17,25,57,90\}$

We want to find out the position of 57, or whether 57 exist or not.

Now the following will be happen:

Binary search

1	2	3	9	11	13	17	25	57	90
a[0]	a[1]	a[2]	a[3]	a[4]	a[5]	a[6]	a[7]	a[8]	a[9]





Question?